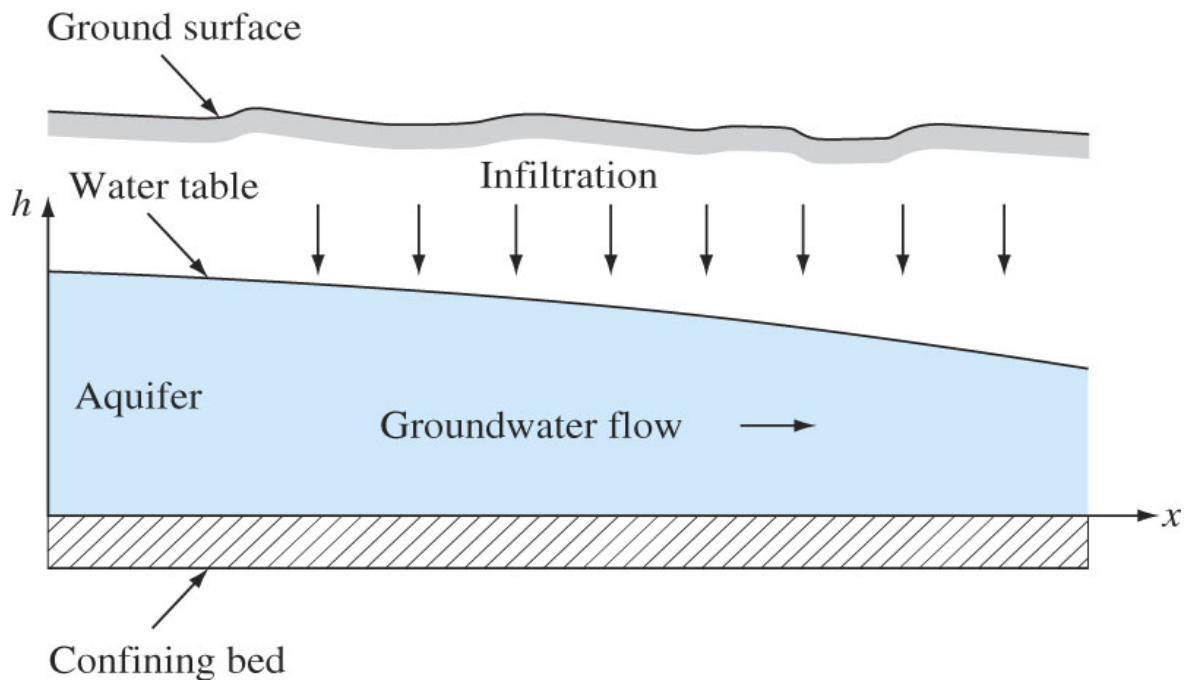


Mini Project (15%) - One dimensional groundwater flow problem

The governing differential equation for steady one-dimensional groundwater flow through an unconfined aquifer shown below can be expressed by

$$\frac{d}{dx} \left(Kh \frac{dh}{dx} \right) + N = 0$$

Where x = distance (m), K = hydraulic conductivity (m/d), h = height of water table (m), and N = infiltration rate (m/d).



- (i) Solve for the height h of the water table from $x = 0$ to 1000 m with $h(0) = 10$ and $h(1000) = 5$ m using (a) MATLAB `bvp4c` function, and (b) Central finite-difference method and compare the solutions in the same plot. Use $K = 1$ m/d and $N = 0.001$ m/d.

If K is constant, then the above equation could be written as a linear equation for h^2 as

$$\frac{K}{2} \frac{d^2(h^2)}{dx^2} + N = 0$$

In this case, we could solve for h^2 with the boundary conditions expressed for h^2 instead of h . An analytical solution could also be obtained by direct integration.

- (ii) A more realistic formulation of the above problem involves heterogeneous hydraulic conductivity $K(x)$ that can vary in space. If $K(x)$ follows a parabolic distribution with $K(0) = 2$,

$K(500) = 1$, and $K(1000) = 0.5$ m/d, determine the coefficients a , b , and c of the quadratic equation $K(x) = ax^2 + bx + c$. Reformulate the problem for this variable $K(x)$ and solve using (a) MATLAB `bvp4c` function, and (b) Central finite-difference method and compare the solutions. All other conditions remain the same as in (i).

In this case also, the equation could be linearized by re-writing in terms of h^2 .

$$K \frac{d^2(h^2)}{dx^2} + \frac{d(h^2)}{dx} \frac{dK}{dx} + 2N = 0$$

- (iii) If the value of N is unknown but the hydraulic head in the middle, $h(500)$, is measured to be 20 m, determine the value of N by using trial and error or by using a nonlinear regression procedure (e.g., MATLAB function `lsqnonlin`). Use the variable K of Part (ii) for this part.
- (iv) Solve the time dependent groundwater flow equation expressed below by using the **ODE approach** for the case given in (ii) for up to $T = 1000$ days using a time step of 1 day.

$$S \frac{\partial h}{\partial t} = \frac{d}{dx} \left(Kh \frac{dh}{dx} \right) + N$$

Assume that storativity $S = 0.1$ and all other values remain the same as in (ii). For the ODE approach, there is no need to linearize the equation. Plot the head profiles (h vs x) for $t = 10$, 100, and 1000 days. You may use linearly interpolated values of the boundary conditions ($h = 10$ to 5 for $x = 0$ to 1000) as the initial condition.